

POCKETSMART AI

Your Smart Budget & Recommendation Assistant

PROJECT REPORT

Submitted for: SmartBridge Project Submission

Course: B.Sc. Computer Science – Semester III

College: Government Arts & Science College

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1. ABSTRACT

PocketSmart AI is an AI-powered budget planning and recommendation web application designed to help users make better decisions while staying within a specified budget.

The application provides three major planning modules: **Home Interior Planner**, **Party Budget Planner**, and **Jewelry Budget Planner**. Users provide their requirements, budget and other relevant details, and the application processes these inputs to generate structured recommendations.

The project uses **Google Gemini** for AI-powered recommendation generation. The Jewelry Planner additionally supports an optional outfit image, allowing Gemini to analyze the outfit's visible colors, style and formality before generating suitable jewelry recommendations.

The application also provides user registration, login, dashboard navigation, recommendation history and budget summaries. SQLite is used for storing user information and recommendation history, while FastAPI manages the backend routes and application logic.

2. INTRODUCTION

Planning purchases and events within a fixed budget can be difficult because users often need to compare different products, services and categories at the same time.

For example, a person planning a home interior may need furniture, lighting and other items while staying within a particular budget. Similarly, a person organizing a party needs to divide the budget between venue, catering, decoration and entertainment.

PocketSmart AI addresses these problems by combining **budget planning with generative AI recommendations**.

Instead of simply displaying a list of products, the application accepts the user's requirements and produces organized recommendations with prices, quantities, category allocations and source platforms.

The project covers three practical domains:

- Home Interior Planning
- Party Planning
- Jewelry Selection

The implemented application follows the SmartBridge project requirements while using FastAPI, Gemini, Jinja2, HTML/CSS, JavaScript and SQLite.

3. PROBLEM STATEMENT

Users often face difficulties when planning purchases or events with a limited budget.

The main problems addressed by PocketSmart AI are:

- Difficulty in distributing a fixed budget across multiple categories.
- Time required to compare different products and services.
- Difficulty choosing products according to personal preferences.
- Need to consider factors such as room type, guest count, occasion and style.
- Difficulty finding suitable jewelry based on an outfit.
- Need to keep recommendations within the available budget.
- Need to access previously generated recommendations.

PocketSmart AI provides a single application where these planning requirements can be entered and converted into structured AI-assisted recommendations.

4. OBJECTIVES

The main objective of PocketSmart AI is to develop a practical AI-assisted budget planning application.

Specific objectives

1. To provide a Home Interior Planner.
2. To provide a Party Budget Planner.
3. To provide a Jewelry Budget Planner.
4. To use Gemini for AI-generated recommendations.
5. To support image-based outfit analysis in the Jewelry Planner.
6. To display prices and budget allocations clearly.
7. To provide source and shopping-platform references.
8. To provide user registration and login.
9. To maintain user sessions.
10. To store recommendation history using SQLite.
11. To provide fallback recommendations when AI output is unavailable or insufficient.
12. To provide a simple and user-friendly interface.

5. PROPOSED SOLUTION

PocketSmart AI provides a common platform containing three specialized planners.

Planner	Main Inputs	Main Output
Home Interior	Budget, room, quantities and preferences	Lighting, fan, furniture and additional suggestions
Party Planning	Budget, guests, event, venue and requirements	Venue, catering, decoration and entertainment
Jewelry Planning	Budget, occasion, style and optional outfit	Outfit analysis, jewelry recommendations and styling tips

PocketSmart AI provides a common platform containing three specialized planners. The three planners share common application features such as authentication, session handling, recommendation history and AI processing.

6. TECHNOLOGY STACK

Technology	Purpose
Python	Application and backend logic
FastAPI	Backend framework and API routing
Uvicorn	Running the FastAPI application
Google Gemini	AI recommendation generation
google-genai	Gemini API integration
Jinja2	Dynamic HTML template rendering
HTML	Web page structure
CSS	User interface design
JavaScript	Client-side interaction
SQLite	User and recommendation-history storage
python-dotenv	Environment configuration
Session Middleware	User session management
VS Code	Development environment
Git & GitHub	Version control and project repository

7. SYSTEM ARCHITECTURE

The application follows a web-based architecture.

User → Frontend → FastAPI Backend → Gemini AI → Recommendation Processing → Result Page

SQLite is used alongside the backend for user information and recommendation history.

Basic workflow

1. User opens PocketSmart AI.
2. User registers or logs in.
3. User accesses the dashboard.
4. User selects one of the three planners.
5. User enters budget and planning requirements.
6. FastAPI receives the submitted information.
7. The appropriate planner logic prepares the AI request.
8. Gemini generates recommendation content.
9. The backend processes the response.
10. Recommendations are displayed in a structured result page.
11. When the user is logged in, the generated plan can be stored in recommendation history.

For the Jewelry Planner, an uploaded outfit image can also be sent to Gemini for image-aware analysis.

8. GOOGLE GEMINI AI INTEGRATION

Google Gemini is the main AI component of PocketSmart AI.

Gemini is used to understand the user's planning requirements and generate domain-specific recommendations.

Gemini responsibilities

- Understand the user's budget.
- Process contextual requirements.
- Generate suitable recommendations.
- Organize recommendations into categories.
- Provide estimated prices.
- Provide styling or planning suggestions.
- Analyze an uploaded outfit image in the Jewelry Planner.

Jewelry image analysis

The Jewelry Planner supports multimodal input.

The user can provide:

- Budget
- Occasion
- Style preference
- Optional outfit image

The uploaded image can be analyzed for visible characteristics such as **color, style and formality**. The resulting analysis is then used as context for jewelry recommendations.

For example, during testing, an outfit was analyzed as having **black and silver colors with a formal/elegant appearance**, after which matching jewelry recommendations were generated.

9. HOME INTERIOR PLANNER

The Home Interior Planner helps users plan interior requirements within a specified budget.

The planner accepts information such as:

- Budget
- Room
- Number of lights
- Number of ceiling fans
- Furniture requirements
- Dining-table requirements
- Additional preferences

The application generates organized recommendations for categories such as lighting, ceiling fans and furniture.

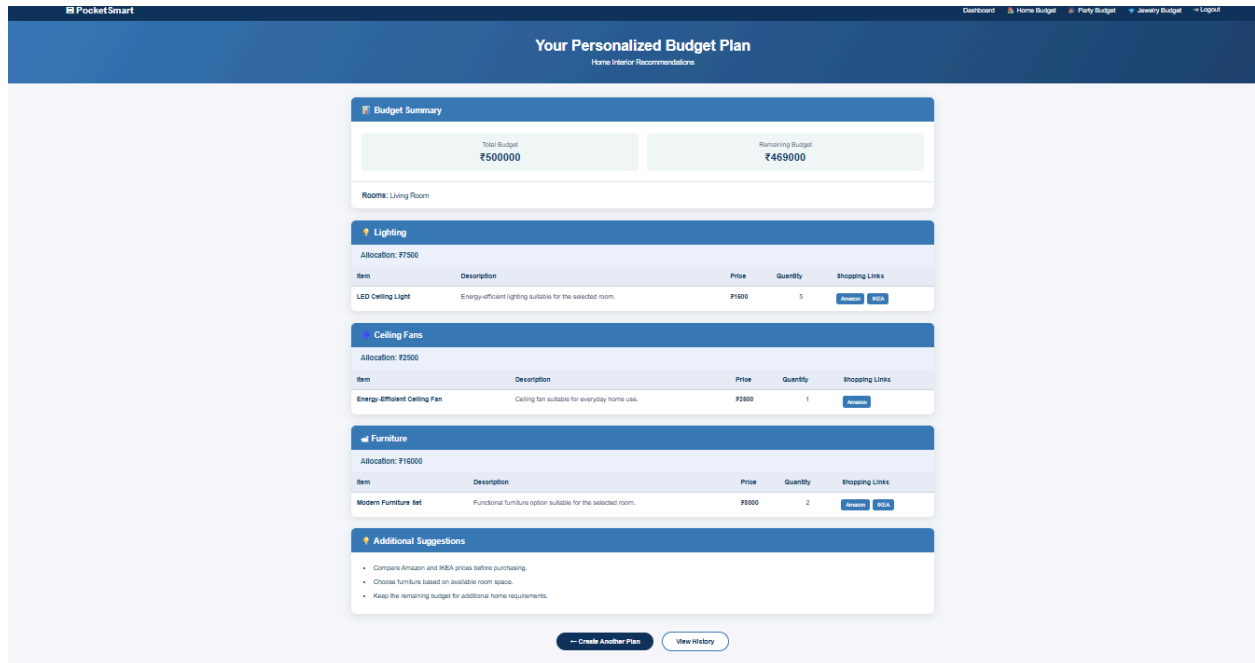
The result page displays the recommended items, quantities, prices, budget information and relevant source/shopping references.

The screenshot displays the 'Home Interior Budget Planner' web application. The top navigation bar includes the 'PocketSmart' logo and links to 'Dashboard', 'Home Budget' (active), 'Party Budget', 'Jewelry Budget', and 'Logout'. The main header area is dark blue with the title 'Home Interior Budget Planner' and the subtitle 'Create a customized budget plan for your dream home interior'.

The main content area is a light gray form with the following sections:

- Budget Details:** A 'Total Budget (₹)' input field with the value '5000'.
- Fixtures & Furniture:** A section with four input fields:
 - 'Number of Lights/Fixtures' with value '5'
 - 'Number of Ceiling Fans' with value '1'
 - 'Number of Furniture Pieces' with value '2'
 - 'Number of Dining Tables' with value '1'
- Rooms to Include:** A section with three checkboxes: 'Living Room' (checked), 'Kitchen' (unchecked), and 'Bedroom' (unchecked).
- Additional Information:** A section with a text area labeled 'Special Requirements or Preferences' containing the placeholder text 'Any specific requirements or preferences...'. A small icon of a pen is visible at the bottom right of the text area.

At the bottom of the form is a blue button labeled 'Generate Recommendations'.



10. PARTY BUDGET PLANNER

The Party Planner helps users organize an event while distributing their budget across different requirements.

The planner collects:

- Total budget
- Guest count
- Event type
- Venue type
- Selected requirements
- Additional requirements

The generated result is divided into categories such as:

- Venue
- Catering
- Decoration
- Entertainment
- Budget summary

[Dashboard](#)
[Home Planner](#)
[Party Planner](#)
[Jewelry Planner](#)
[History](#)
[Logout](#)

11. JEWELRY BUDGET PLANNER

The Jewelry Planner is designed to recommend jewelry according to the user's budget, occasion and style.

The user can provide:

- Budget
- Occasion
- Style preference
- Optional outfit image

The result contains:

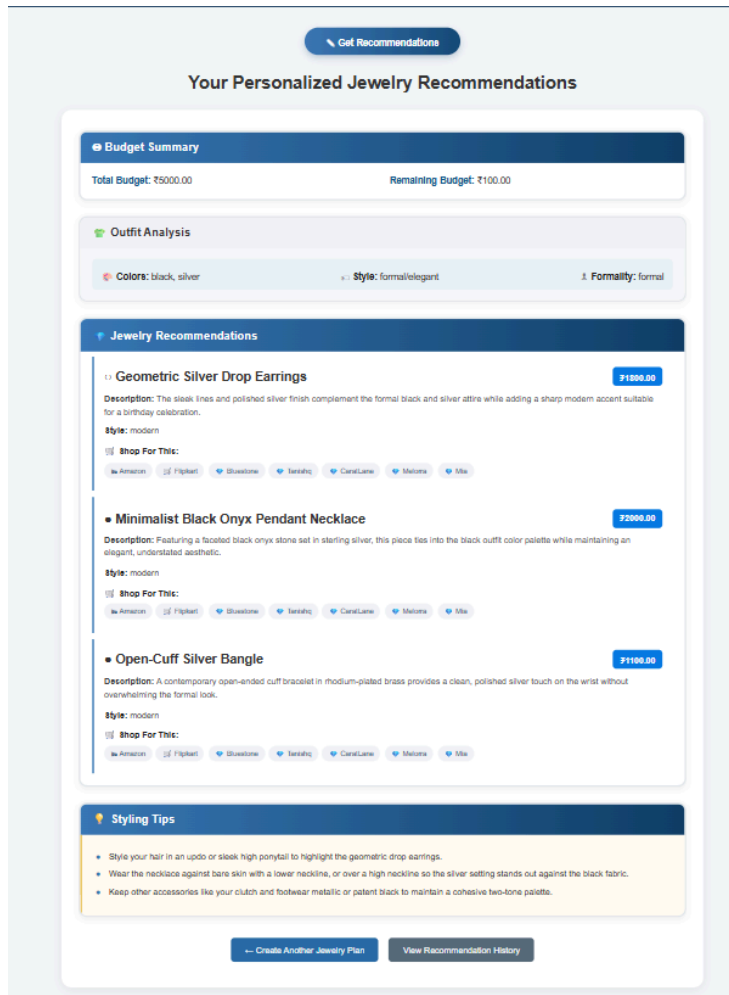
- Outfit analysis
- Jewelry recommendations
- Estimated prices
- Shopping/source platforms
- Styling tips

This makes the Jewelry Planner different from the other planners because it supports **multimodal AI input using both text and an optional image**.

The screenshot displays the 'Jewelry Budget Planner' web application. The interface features a dark blue header with the 'PocketSmart' logo and navigation links: Dashboard, Home Budget, Party Budget, Jewelry Budget (active), History, and Logout. Below the header, the title 'Jewelry Budget Planner' is centered, with a subtitle 'Get AI-powered jewelry recommendations within your budget for any occasion'. The main content area is a light blue card with the following sections:

- Budget Details:** A form with 'Total Budget (₹)' set to 5000.
- Occasion & Preferences:** A dropdown menu for 'Occasion' set to 'Birthday'.
- Style Preferences:** A text area for 'Describe your style preferences, materials, colors, etc.' with a small icon for adding more details.
- Upload Outfit Image:** A section with a 'Choose File' button, a file name 'party dress.jpeg', a preview image of a woman in a black dress, and a 'Remove Image' button.

At the bottom of the card is a blue button labeled 'Get Recommendations'. The footer of the page contains the copyright notice '© 2020 PocketSmart | Smart Budget Solutions'.



12. USER REGISTRATION AND LOGIN

PocketSmart AI includes an authentication system.

New users can create an account by providing their name, email and password.

Existing users can log in using their registered credentials.

After successful login, the user is taken to the dashboard.

The application uses session management to maintain the logged-in user's information during the application workflow.

PocketSmart

AI-Powered Budget Planning

Create Your Account

Username

 Choose a username

Email

 Enter your email

Password

 Create a strong password

Confirm Password

✓ Confirm your password

 + Create Account

Already have an account? [Sign In](#)

PocketSmart

AI-Powered Budget Planning

Welcome Back

Username

 Enter your email

Password

 Enter your password

[Forgot Password?](#)

Sign In →

Don't have an account? [Create Account](#)

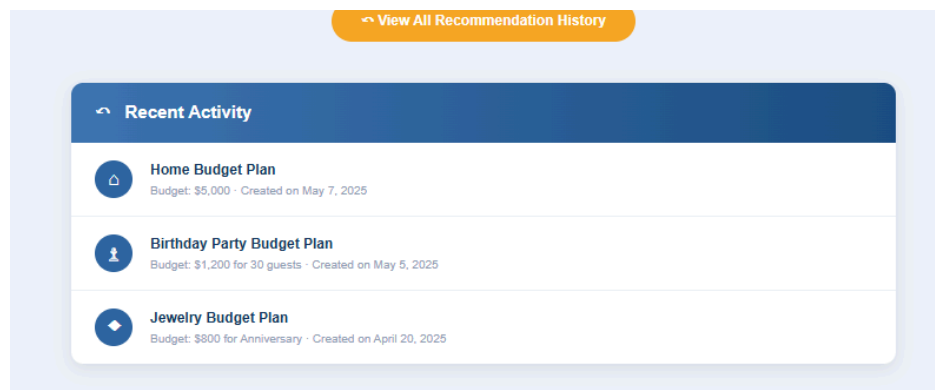
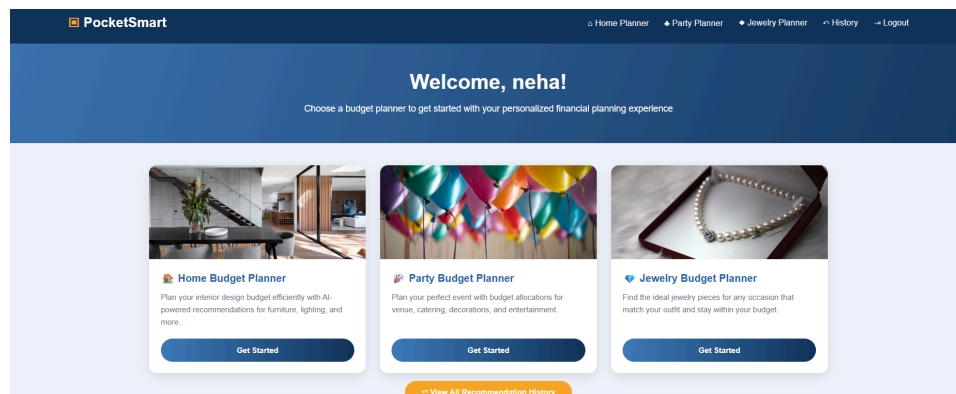
13. USER DASHBOARD

The dashboard acts as the central area for logged-in users.

It provides access to:

- Home Interior Planner
- Party Planner
- Jewelry Planner
- Recommendation History
- Recent Activity
- Logout

The dashboard provides a simple way for users to move between the major functions of the application.



14. RECOMMENDATION HISTORY

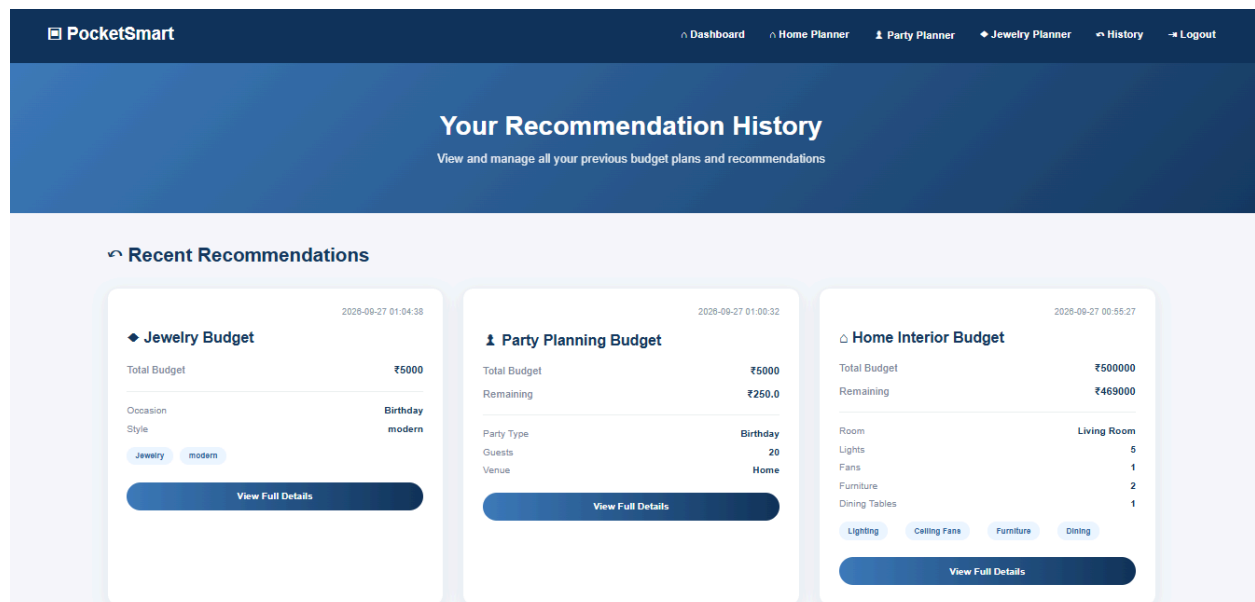
PocketSmart AI provides a Recommendation History feature.

When a planning request is generated while the user is logged in, the relevant information can be stored in SQLite.

The history page allows users to review previous planning activities, including information such as:

- Planner type
- Budget
- Date/time
- Planner-specific details
- Recommendation information

This helps users revisit their previous planning results without starting from the beginning.



15. USER INTERFACE

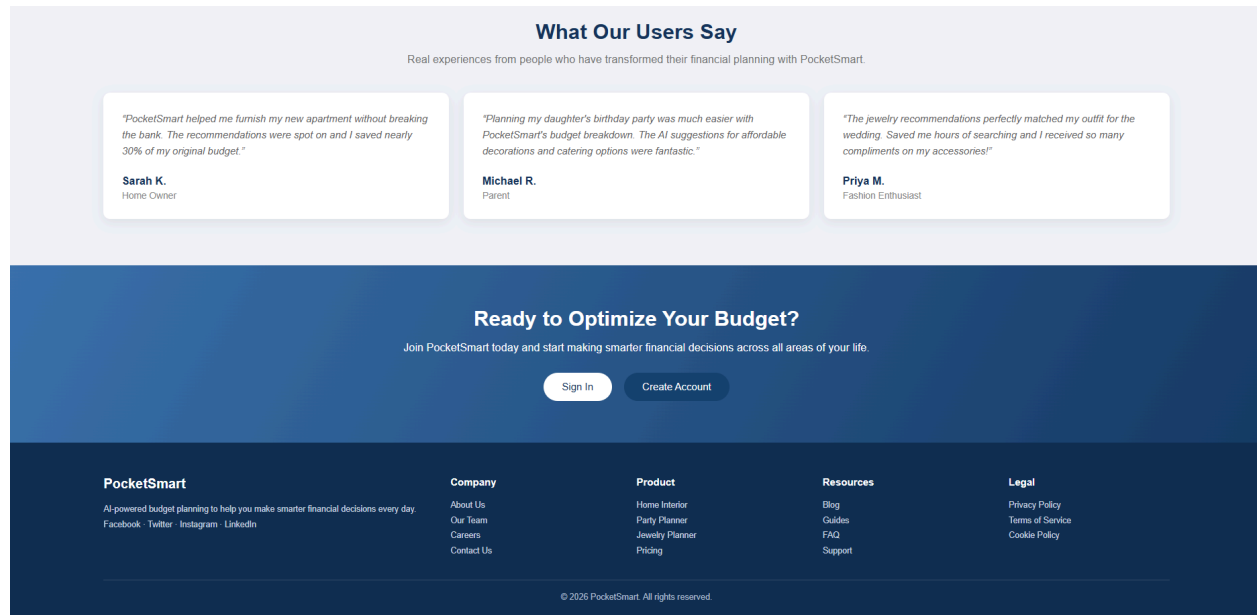
The application contains several connected pages:

- Home
- Testimonials and Footer
- Register
- Login
- Dashboard
- Home Interior Planner
- Home Interior Recommendations
- Party Planner
- Party Recommendations
- Jewelry Planner
- Jewelry Recommendations
- Recommendation History

The interface uses HTML and CSS with Jinja2 templates to display dynamic information returned by the backend.

The result pages use organized cards and sections so that users can easily understand recommendations, prices and budget information.





16. BUDGET MANAGEMENT

Budget management is one of the main features of PocketSmart AI.

The application considers the user's available budget while generating recommendations.

For the Party Planner, the budget is divided among categories such as venue, catering, decoration and entertainment.

For Home Interior and Jewelry planning, recommendations include prices and a budget summary.

The purpose is to help users understand how their available money can be distributed rather than simply receiving an unstructured list of suggestions.

17. DATABASE AND SESSION MANAGEMENT

SQLite is used as the local database for the application.

The database stores:

- User information
- Recommendation history
- Planner details
- Recommendation data
- Creation timestamps

Session middleware is used to maintain the logged-in user's information.

This allows features such as personalized dashboard access and user-specific recommendation history.

18. FALLBACK HANDLING

AI services can sometimes be temporarily unavailable, rate-limited or return insufficient information.

To handle this situation, PocketSmart AI includes fallback/default recommendation logic.

Therefore, the application can still provide a controlled recommendation result when the AI service does not return enough usable information.

This approach is also consistent with the SmartBridge requirement that permits fallback/default recommendations when AI output is insufficient.

19. TESTING AND VALIDATION

The application was tested across the major workflows.

Test Area	Test Performed	Expected Result
Home Planner	Enter budget and room requirements	Home recommendations generated
Party Planner	Enter budget, guests and event details	Category-wise party recommendations generated
Jewelry Planner	Enter budget, occasion, style and outfit image	Outfit analysis and jewelry recommendations generated
Registration	Create user account	Account created
Login	Enter valid credentials	Dashboard displayed
Dashboard	Navigate between planners	Correct pages opened
History	Generate plans while logged in	Plans stored in history
AI fallback	Handle insufficient AI response	Default/fallback recommendations available

The three planner workflows and their resulting pages were tested and captured for the project report.

20. SMARTBRIDGE MILESTONE MAPPING

SmartBridge Area	PocketSmart AI Implementation
Milestone 1 – Gemini Initialization	Gemini API configuration and AI testing
Milestone 2 – Core Functionality	Planner-specific prompts, budget processing and image analysis
Milestone 3 – Backend	FastAPI routes, authentication, sessions, history and recommendation details
Milestone 4 – UI Development	HTML/CSS/Jinja2 pages and planner/result interfaces
Milestone 5 – Testing & Optimization	Planner testing, budget checks, UI testing and fallback handling

The implemented project covers the major functional areas described in the SmartBridge specification.

21. LIMITATIONS

The current project has some practical limitations:

- Platform names shown in recommendations are source/shopping references rather than guaranteed live inventory.
- AI-generated prices are estimates and should not be considered guaranteed real-time prices.
- Recommendation quality depends on the Gemini service response.
- The project is intended as an academic/demo application rather than a production e-commerce platform.

22. FUTURE ENHANCEMENTS

Future versions of PocketSmart AI could include:

- Live product and service APIs.
- Real-time price and availability checking.
- More shopping and service platforms.
- Advanced user preference profiles.
- More accurate recommendation ranking.
- Improved budget optimization.
- More detailed image analysis.
- Mobile application support.
- Secure password hashing and stronger production authentication.
- More planning categories.

23. PROJECT OUTCOME

The completed PocketSmart AI application provides a working AI-assisted planning workflow from **user input** → **budget processing** → **AI recommendation** → **structured result** → **history**.

The project successfully brings together three planning domains:

Home Interior + Party Planning + Jewelry Selection

with common features including authentication, dashboard navigation, recommendation history and budget-aware recommendations.

24. CONCLUSION

PocketSmart AI demonstrates how generative AI can be integrated into a practical budget-planning web application.

The project combines **FastAPI**, **Google Gemini**, **Jinja2**, **HTML/CSS**, **JavaScript** and **SQLite** to create a single platform for different everyday planning needs.

The Home Interior Planner helps organize interior requirements, the Party Planner distributes an event budget across important categories, and the Jewelry Planner provides occasion- and style-based recommendations with optional outfit-image analysis.

The authentication system, dashboard and recommendation history provide a connected user experience, while fallback handling helps maintain functionality when AI output is unavailable or insufficient.

Overall, PocketSmart AI provides a practical demonstration of using generative AI for **budget-aware, personalized recommendations**

25. PROJECT DELIVERABLES

The project deliverables include:

- Working PocketSmart AI web application
- Home Interior Planner
- Party Budget Planner
- Jewelry Budget Planner
- Gemini AI integration
- Authentication and dashboard
- Recommendation History
- Project source code
- GitHub repository
- Project demonstration video
- Project report with screenshots

This is the kind of report you should submit: **it explains your finished project, its features, architecture, implementation and results**, rather than teaching the evaluator how to build it.